



LinearTrade

Evidence-first AI for NSE equity research

Seven agents argue the case.
Plain code makes the call.



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Agenda

- | | | | |
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Most retail backtests flatter the strategy

Before an idea is worth trading, it has to clear three bars.

₹237.82

Charges on a ₹1 lakh round trip at Zerodha. Most of it is STT.

1 min

From idea to order for a typical retail investor, with no checks in between.

4

Silent biases in a typical home-made backtest. None raise an error.



Four silent biases, four mechanical fixes

01 Same-day fill

Buys at the close that produced the signal.

✓ Fill at the next day's open

02 Charges left out

STT, stamp duty and GST are ignored.

✓ Dated, statutory charge schedules

03 Survivorship

Today's NIFTY 50 is tested on the past.

✓ Index membership by date

04 Unadjusted prices

A 1:1 bonus reads as a 50% crash.

✓ Adjust before any indicator



Built for three people

Each one asks the system a different question.



Researcher

"Is this stock worth my attention?"

Research desk, debate, backtests



Operator

"What did it do today, and why?"

Scheduler, cockpit, portfolio monitor



Reviewer

"Can I rebuild this decision?"

Journal, rejected signals, scorecard



Agents recommend. Code decides.

Language models are good at reading and arguing. They are not trusted with money.

Agents

- 1 Read charts, fundamentals and news
- 2 Argue bull against bear
- 3 Propose BUY, SELL or HOLD

Hard boundary

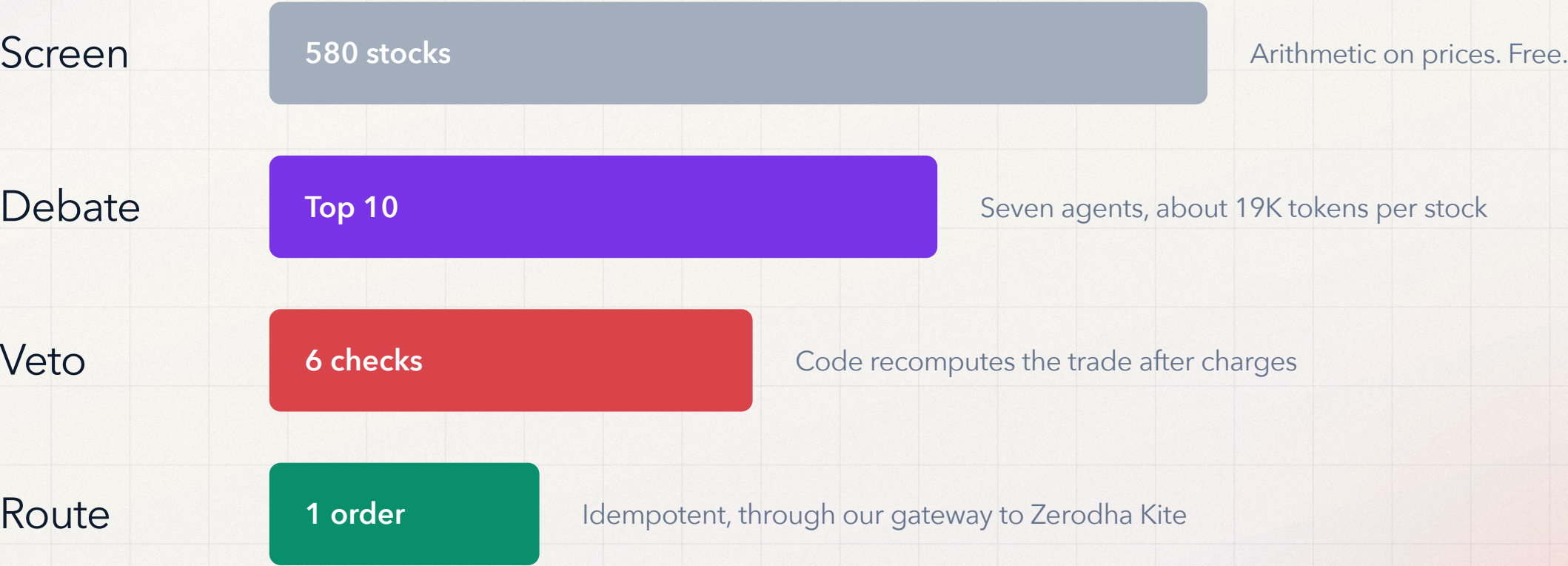
Code

- 1 Recompute every number after charges
- 2 Apply risk limits and the kill switch
- 3 Size and route the order



Four stages, cheapest first

Only one stage spends tokens, and it runs on the fewest stocks.





1.27 million daily bars, each one dated

Every backtest sees only what was knowable on that day.

580

NSE stocks

2015-26

Years covered

562

Liquid-200 members, by date

4

Dated charge schedules

Trading store

Used for trades

Parquet + DuckDB. Feeds every backtest and every debate.

Research lake

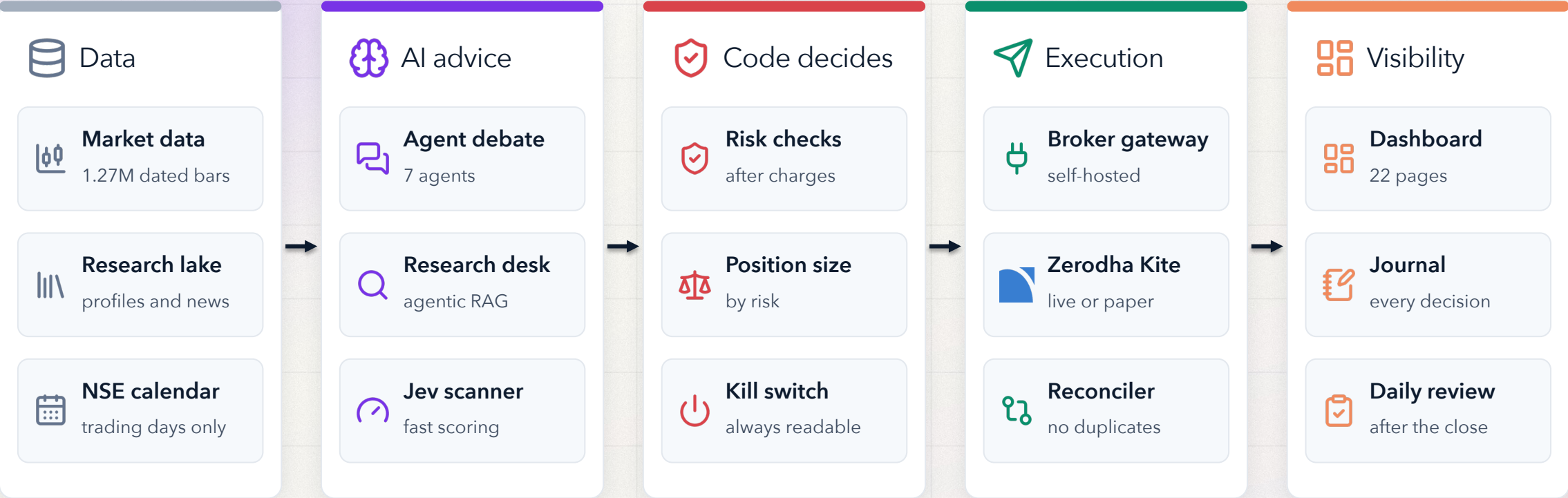
Never places an order

284 company profiles, 2,759 news items. Read-only.



How the whole system fits together

Data comes in, AI agents advise, code decides, the broker executes.

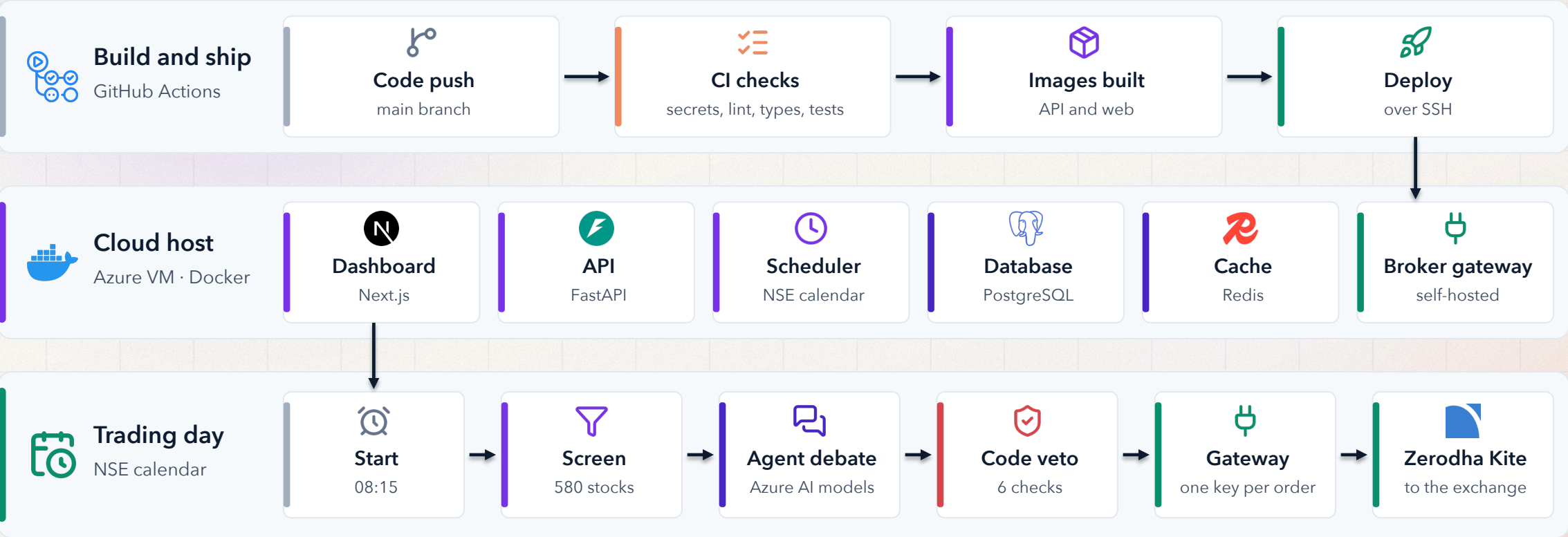


 A scheduler on the NSE calendar runs every step, and every step is logged



From cloud to live order

How the code ships, where it runs, and what happens on every trading day.



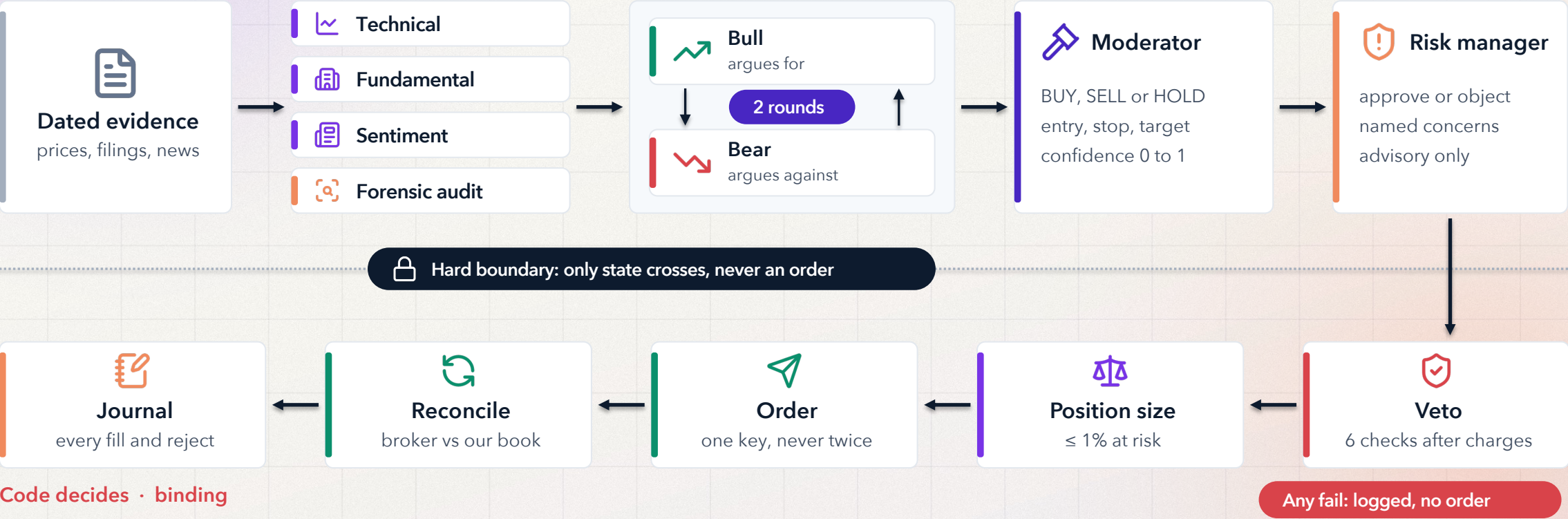
⦿⦿⦿ No public ports: the dashboard is reachable only over a private Tailscale network



Inside the agent debate

What each step receives, and what it hands to the next.

AI advice · LangGraph





Any check can say no

From a live run: the model liked the trade, and the code caught the cost it missed.

Reward to risk

1.85 → 1.76

Before costs

Model: approve

After 12 bps of costs

Code: blocked

The minimum is 1.8, measured after charges on both legs.

Six checks, cheapest first

- 1 Stop on the correct side
- 2 Reward to risk ≥ 1.8 after costs
- 3 Expected gain $\geq 3 \times$ round-trip cost
- 4 Confidence floor
- 5 Stock, sector and daily limits
- 6 Kill switch readable and off

Sized by risk: at most 1% of the account lost at the stop.



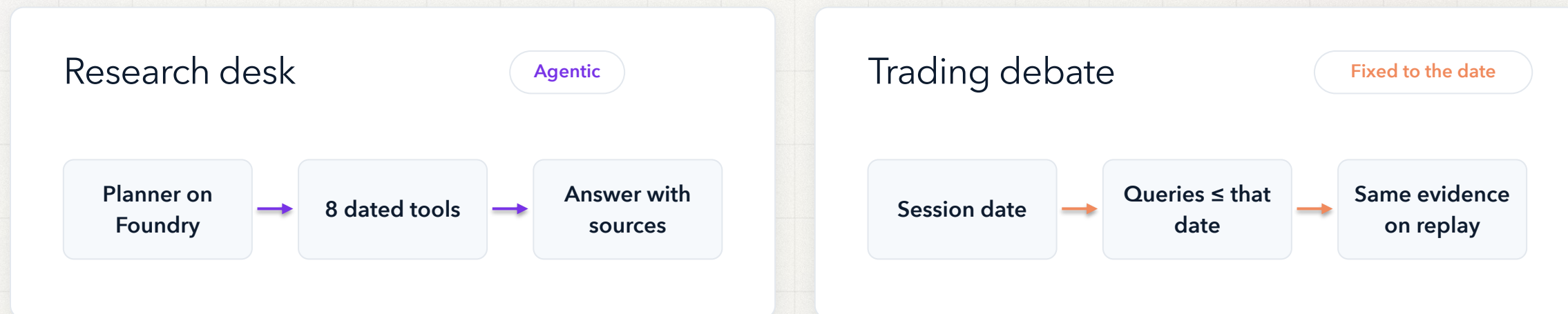
Built on TradingAgents, hardened for money

Same layout: analysts, bull against bear, moderator. Different where money is at stake.

	TradingAgents	LinearTrade
Agent output	Free text, parsed by a second call	A typed schema from every agent
Final say	A language model as risk judge	A veto written in code
Costs	Left out of reward to risk	Both legs, after charges
Prompts	Strings inside the code	Versioned files in git
A failing agent	Stops the whole run	Recorded by name; the run continues

Every number carries a date

Two retrieval paths. Neither can see data published after the day in question.



97%

Lake agrees with NSE data (1,228 checks)

7 days

Oldest data an answer may use

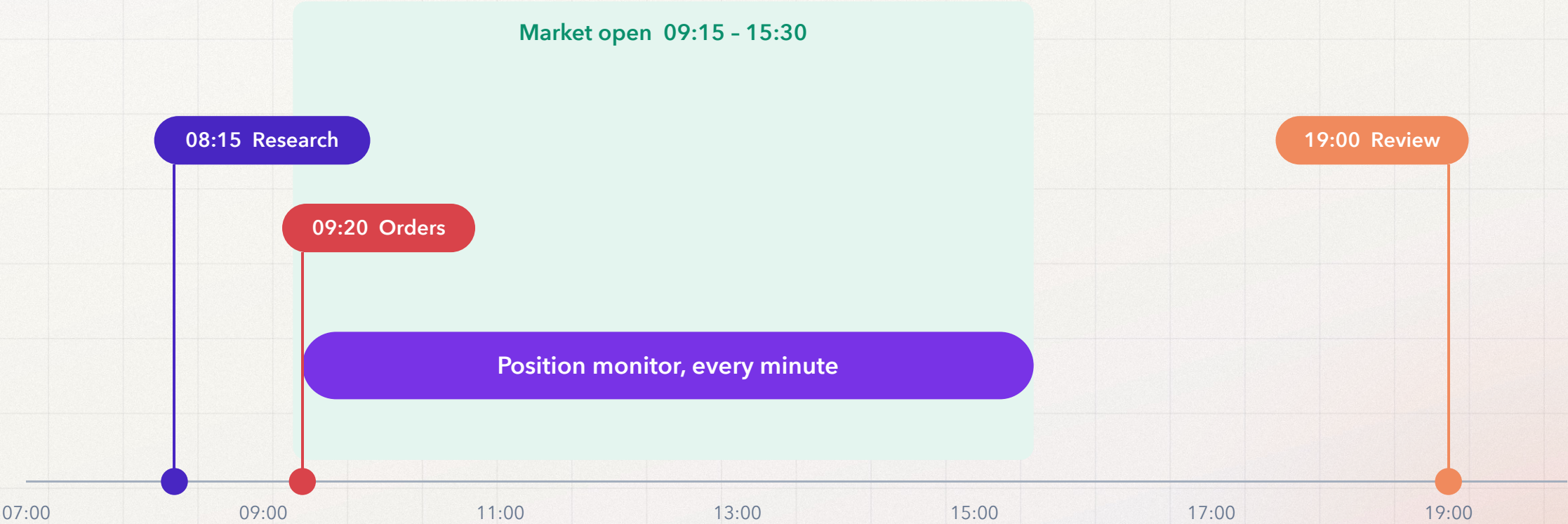
6 rounds

Most tool loops the planner may run



One trading day, on the NSE calendar

Nothing runs on an exchange holiday. A job more than 30 minutes late is skipped.





The stack

One reason for each choice. Versions as installed on 3 Oct 2026.



AGENTS

LangGraph 1.2

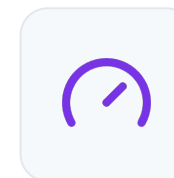
Debate rounds are counted



MODELS

Foundry · gpt-5-mini

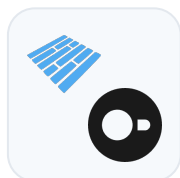
One endpoint, any model



SCORING

Jev System One

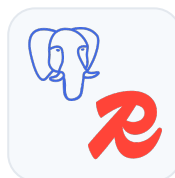
Probabilities, not prose



MARKET DATA

Parquet + DuckDB

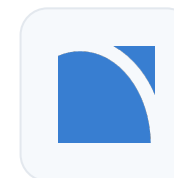
Fast scans over 1.27M bars



STATE

Postgres 16 · Redis 7

Transactions plus job locks



BROKER

Zerodha Kite

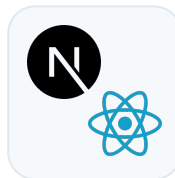
Self-hosted, paper or live



SERVICE

FastAPI · APScheduler

Jobs follow the NSE calendar



INTERFACE

Next.js 16 · React 19

22 pages on one API



QUALITY

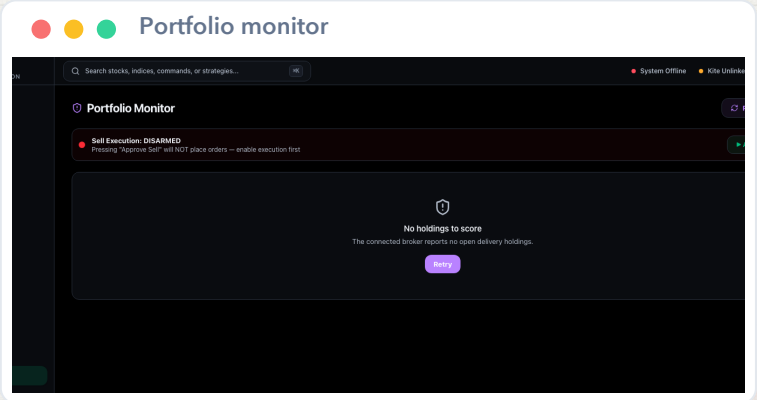
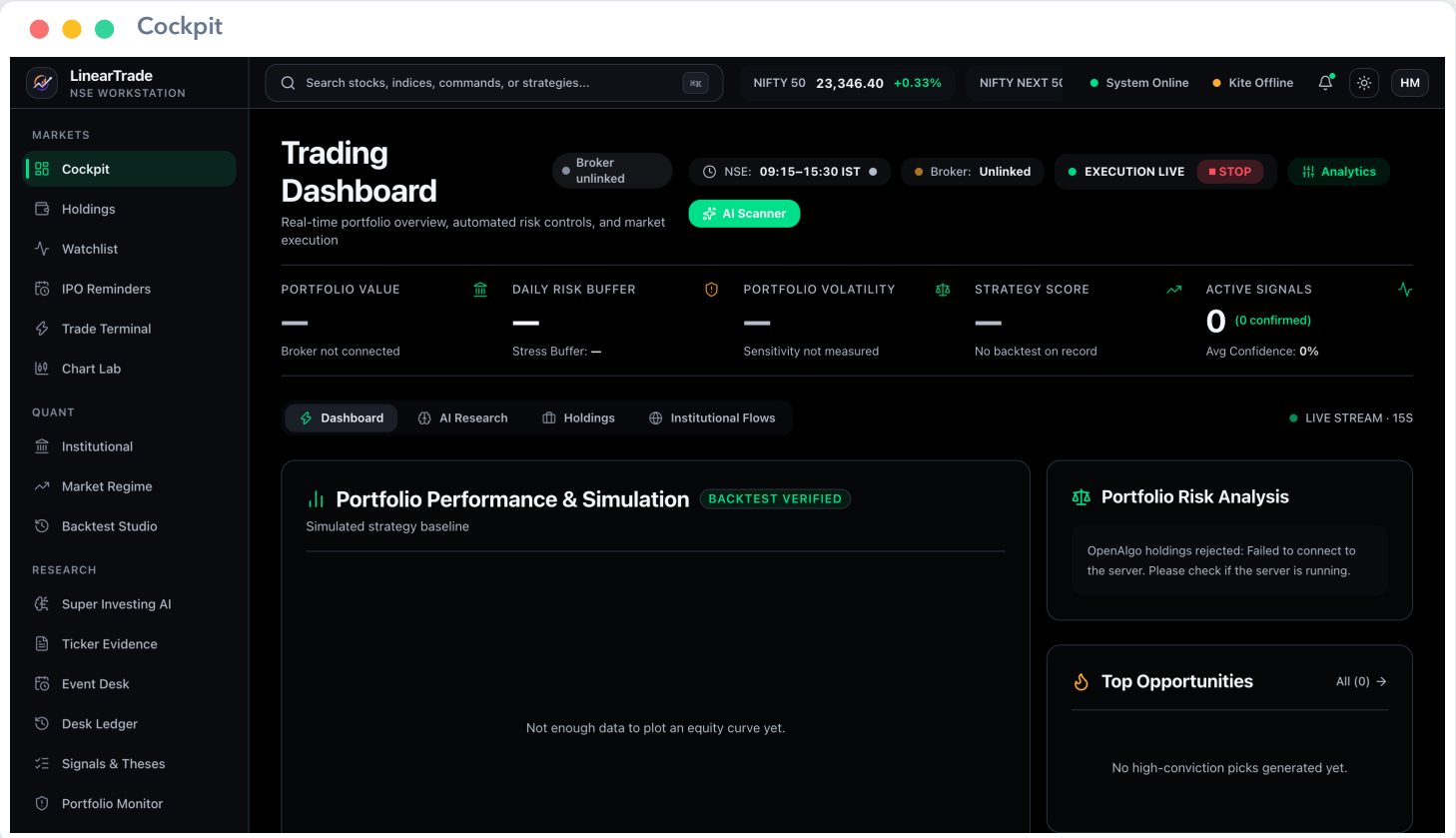
ruff · mypy · pytest

Strict types, layer rules in CI



What it looks like

When something can't be measured, the screen says why instead of showing a number.



Jev: a second opinion in 280 ms

Two fixed questions about volatility. Probabilities out. Code still sizes the trade.



What Jev is asked

Is the market calm?

Nifty volatility below its 100-day median, no crash

Is this stock calm?

Its own volatility below median, no big move today

Last scan, data to 1 Oct 2026

Capital protected

The index fell 6.2% in 20 days. Jev read the market as unsettled and kept every NSE position at zero.

Where Jev is used

Live scanner every scan

Inside the platform same code veto

Calibration judged on outcomes



We judge every strategy by efficiency

Return on data it never saw, divided by return on the data it was fitted to.

Efficiency

unseen return
÷ fitted return

0.95 best strategy

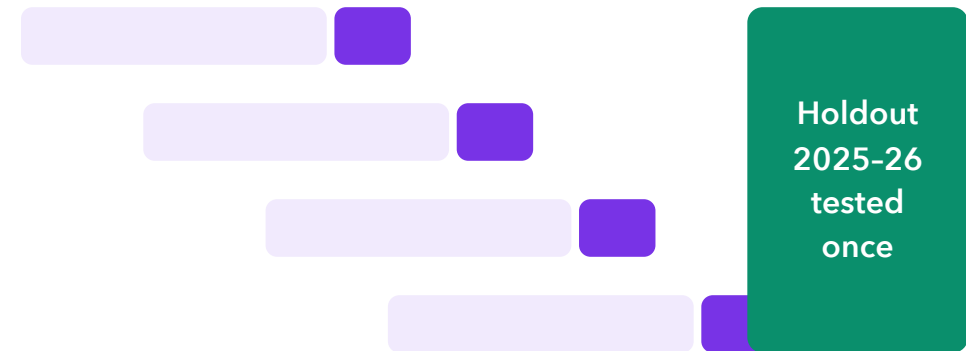


0 overfit

1 generalises

Also reported: Sharpe, drawdown, deflated Sharpe, probability of overfitting.

Walk-forward, then one untouched holdout



24 months to fit, 6 to test, 20 windows

Every fill at the next day's open, with dated charges and index members by date.



Strategies that hold up on unseen data

Study 1: tested across 20 walk-forward windows, after real Indian charges.

Best strategy: momentum on a liquid top-200

+134.22%

Return on data the strategy never saw

0.95

Efficiency: unseen return kept pace with fitted

20

Walk-forward windows, each tested out of sample

Study 2: frozen candidate, untouched holdout

Candidate



NIFTY 50



Ahead of the index by 9.75 points on 2025-26 data it never saw.



Jev's questions pick better trades

Study 3: unseen data. A win is a 3 ATR gain before a 1.5 ATR loss, within 20 sessions.

Hit rate of GO calls, holdout

31% vs 22%
base rate

+0.20R

Average result per GO trade on the holdout

Edge of GO calls over taking every trade, R per trade

Main holdout
280 samples



12 stocks never used in design
720 samples



All unseen data
1,300 samples





Objectives delivered



Trustworthy data

Index members by date and verified 2026 charge schedules



Honest measurement

Walk-forward, CPCV, deflated Sharpe and an untouched holdout



Model kept in check

A veto in code, outside the agent graph



Grounded research

Live answers matched the dated snapshot



The model advises. The code decides.

An agentic research system that checks its own work before any money moves.

+134.22%

Return on unseen data, best strategy

0.95

Efficiency on unseen data

+0.10R

Jev edge across 1,300 unseen samples

Built: a seven-agent debate, a dated research desk, a veto in code, and the Jev scanner.



Thank you

Questions welcome, especially about the numbers.

github.com/HarishM-AIML-A-58/Trading-Agent